

Post-FEC Performance Prediction Using Distribution Transform Functions Under Non-Stationary Conditions

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Abstract Static pre-FEC thresholds fail to guarantee requirements under non-stationary conditions. We propose a transform function leveraging pre-FEC distributions for accurate post-FEC estimation, simplifying to specific distribution moments at high SNR. Experimental validation confirms superior accuracy, preventing significant link margin monitoring errors in dynamic fiber environments. ©2026 The Author(s)

Introduction

As optical systems transition into the Terabit era, accurate performance prediction has become essential [1, 2]. With signal-to-noise ratio (SNR) margins becoming critically small, unforeseen performance degradations can lead to catastrophic failures in high-capacity applications, such as data center interconnects [3, 4]. Consequently, robust reliability modeling is of paramount importance to the optical communications community.

For decades, static pre-Forward Error Correction (pre-FEC) Bit Error Rate (BER) thresholds have been employed to guarantee post-FEC performance across various standards [5–7]. These static limits are strictly valid under Additive White Gaussian Noise (AWGN) conditions, where error counts follow a classical binomial distribution over a fixed frame length. However, non-stationary conditions in modern optical systems cause the BER distribution to deviate significantly from the binomial model, rendering static thresholds incapable of guaranteeing post-FEC integrity.

Time-varying impairments in modern coherent systems introduce non-stationary behavior in BER sequences. Notably, fiber nonlinearity [8], timing jitter [9], and Equalization-Enhanced Phase Noise (EPPN) [10–12] can induce heavy-tailed pre-FEC BER distributions. Even at an identical mean BER, FEC performance degrades significantly when the error distribution exhibits these heavy-tailed characteristics. This degradation highlights why traditional point-estimate thresholds fail to provide accurate performance guarantees.

While prior research has explored FEC performance prediction, existing models are often restricted to AWGN conditions [13, 14] or rely on semi-analytical approaches that are computationally intensive [15, 16], limiting their applicability in real-time transponders. To address this gap, we propose a Distribution Transform Function (DTF) method. This approach accurately predicts post-FEC BER under non-stationary conditions

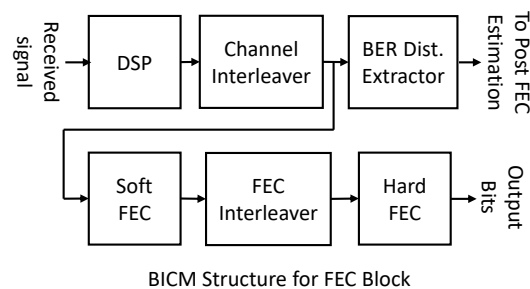


Fig. 1: FEC decoding block based on bit-interleaved coded modulation (BICM), used as an example in this work.

by mapping the input error distribution directly to the output performance. We demonstrate that this method simplifies to calculating a specific moment of the BER distribution in high-SNR regions.

In this work, we first establish the system model and the corresponding experimental setup. We provide a rigorous mathematical justification for the DTF framework and describe the methodology for adapting it to various code structures. As a practical implementation, we evaluate a concatenated FEC scheme comprising a Soft-Decision (SD) BCH(126, 110, 2) code using Chase decoding [17] and an RS(544, 514, 10) code (KP4). Our experimental results—spanning AWGN, non-stationary, nonlinear, and jitter-dominated regimes—demonstrate a robust agreement between the predicted and measured performance.

System Model and Experimental Setup

In this work, we focus on the BER evolution through the FEC decoding block, as illustrated in Fig. 1. We define the frame BER as the ratio of errored bits to the total number of bits over a fixed frame length. The frame BER is treated as a random variable characterized by a specific probability distribution. This distribution is obtained via the BER distribution extractor block shown in Fig. 1, utilizing either feedback from the FEC output or dedicated pilot symbols for error counting. The proposed post-FEC prediction framework re-

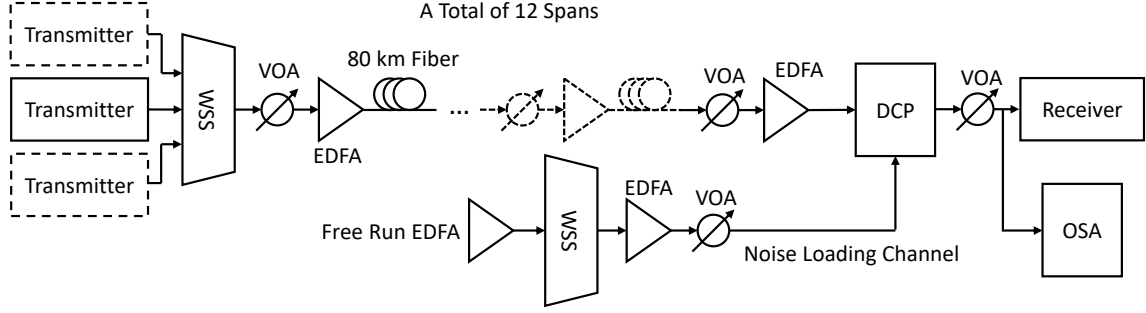


Fig. 2: Experimental setup used for BER statistics extraction and BER prediction validation. The receiver is capable of returning 16,384 samples of error counts calculated over a frame length of 655,360 bits. This diagnostic function, called FastBER, enables extraction of the frame BER distribution.

lies on access to this frame-wise BER distribution as its primary input.

The experimental validation is conducted using the setup depicted in Fig. 2. The transmitter comprises a 200G transponder utilizing dual-polarization 16-QAM (DP-16QAM) modulation. The signal is launched into standard single-mode fiber (SSMF) with a total transmission length of 960 km in the C-band (1550 nm). The link is configured with 12 spans of 80 km each. Every span terminates with a Variable Optical Attenuator (VOA) followed by a fixed-gain Erbium-Doped Fiber Amplifier (EDFA) with a gain of 20 dB (to compensate for span loss) and a noise figure of 5 dB. Pre-FEC BER distribution data is collected using the transponder's *FastBER* diagnostic function, which provides 16,384 frame BER samples. Each sample is calculated over a frame length of 655,360 bits, serving as the statistical basis for our DTF performance prediction model.

Methodology

In this section, we introduce the proposed DTF framework and provide a mathematical justification for its application in performance prediction.

We define the post-FEC BER, denoted as $b_p = \mathbb{E}[b_{p,f}]$, as the ergodic average of the error rate across a sequence of decoded frames, where $b_{p,f}$ is the frame-wise BER at the FEC output calculated over a frame length L . By applying the Law of Total Expectation, we express the post-FEC BER as a function of the pre-FEC error distribution:

$$\begin{aligned} b_p &= \sum_{i=0}^L \mathbb{E} \left[b_{p,f} \mid b_{i,f} = \frac{i}{L} \right] \cdot \mathbb{P} \left(b_{i,f} = \frac{i}{L} \right) \\ &= \sum_{i=0}^L \text{DTF} \left(\frac{i}{L} \right) \cdot \mathbb{P} \left(b_{i,f} = \frac{i}{L} \right) \end{aligned} \quad (1)$$

where $\mathbb{P}(b_{i,f} = \frac{i}{L})$ is the probability of having exactly i errors in an input frame of length L . The term $\text{DTF}(\frac{i}{L}) = \mathbb{E}[b_{p,f} \mid b_{i,f} = \frac{i}{L}]$ represents the average output BER conditioned on exactly i input errors. This function transforms the input error distribution into the expected post-FEC performance

according to the code weight enumerator and decoding algorithm.

Given a sufficiently large frame length L , the FEC decoding process becomes independent of the error patterns in adjacent frames. Consequently, the DTF is an intrinsic characteristic of the FEC code structure and decoder implementation, remaining invariant to underlying channel conditions. This allows the DTF method to be generalized from standard AWGN channels to non-stationary scenarios where the pre-FEC BER distribution deviates significantly from the binomial model.

While an analytical derivation of the DTF is possible for algebraic codes such as BCH and RS, we propose a simplified power-law approximation. In the high-SNR waterfall region (typically $b_p < 10^{-7}$), the relationship between input and output BER follows a power-law behavior [18]:

$$\text{DTF}(b_{i,f}) \approx \theta \cdot (b_{i,f})^\gamma \quad (2)$$

where θ and γ govern the position and slope of the waterfall curve for a specific FEC mechanism at a fixed frame length. These parameters can be empirically determined using an AWGN calibration setup. By measuring two sets of pre/post-FEC BER pairs $(b_{i,1}, b_{p,1})$ and $(b_{i,2}, b_{p,2})$ under AWGN conditions, θ and γ are extracted by solving the following system, noting that the pre-FEC BER follows a binomial distribution:

$$b_{p,j} = \sum_{i=0}^L \theta \left(\frac{i}{L} \right)^\gamma \binom{L}{i} b_{i,j}^i (1 - b_{i,j})^{L-i}, j = 1, 2 \quad (3)$$

Substituting (2) into (1), the post-FEC prediction becomes:

$$b_p \approx \theta \cdot \mathbb{E}[(b_{i,f})^\gamma] \quad (4)$$

This formulation reveals a critical insight: to guarantee post-FEC performance under non-stationary conditions, the system must monitor the γ -th moment of the pre-FEC BER distribution rather than relying on a static mean threshold.

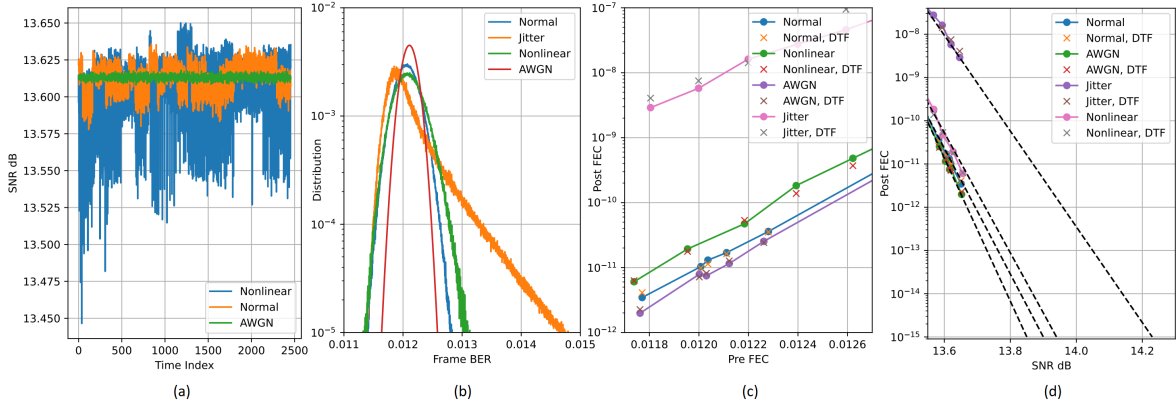


Fig. 3: Experimental results: (a) SNR fluctuation over time, (b) frame BER under different scenarios with the same mean BER, (c) FEC performance, and (d) SNR requirements to guarantee FEC performance.

Experimental Results and Discussion

The first experiment characterizes the non-stationary behavior of the link. We measured the SNR at launch powers of 4.3 dBm and 6.9 dBm for the linear and nonlinear regimes, respectively. These powers were selected to yield an identical mean pre-FEC BER of 1.21×10^{-2} . Measurements were averaged over blocks of 1.6×10^7 symbols, with results presented in Fig. 3(a). For comparison, the expected behavior of a static AWGN channel is also plotted. The observed SNR in the nonlinear case exhibits temporal fluctuations that significantly exceed the AWGN reference. Furthermore, the nominal (linear) case also demonstrates higher fluctuations compared to the AWGN baseline. We attribute these time-varying dynamics to channel state fluctuations and the adaptive nature of the DSP algorithms, which cause error events to be non-identically distributed.

In the second experiment, we utilized the collected transponder data (FastBER) to extract pre-FEC frame BER distributions under various conditions, as illustrated in Fig. 3(b). It is important to note that all evaluated scenarios share an identical mean pre-FEC BER of 1.21×10^{-2} . For comparison, we plotted the theoretical binomial distribution—corresponding to the same frame length and mean BER—as a reference for the AWGN channel. Even under nominal operating conditions, the measured distribution is notably wider than the AWGN baseline. In the jitter-dominated scenario, a pronounced heavy tail appears on the right side of the distribution, indicating bursty error behavior. While all measured curves in Fig. 3(b) maintain the same mean pre-FEC BER, their distributional differences arise from non-stationary channel conditions, ultimately resulting in significantly divergent post-FEC performance.

The subsequent experiment evaluates post-FEC BER across a range of channel powers and corresponding SNR levels. Fig. 3(c) illustrates the post-FEC BER as a function of the mean pre-FEC BER. For a fixed mean pre-FEC BER, the

system exhibits divergent post-FEC performance across the AWGN, nominal, nonlinear, and jitter-dominated regimes. To predict this behavior, we applied the DTF method using Eq. (4) with parameters $\theta = 1.4979 \times 10^{99}$ and $\gamma = 57.477$, calibrated for the concatenated SD-BCH + KP4 scheme at a target post-FEC BER of 10^{-10} . As shown, the DTF predictions align closely with the experimental measurements across all scenarios, confirming the model's accuracy in non-stationary environments. Notably, these results demonstrate that the FEC decoder yields different output performance for the same mean input BER, confirming that static pre-FEC thresholds are insufficient for guaranteeing post-FEC reliability.

In the final experiment, we extrapolated the post-FEC performance curves to a target BER of 10^{-15} to assess link margin requirements. As illustrated in Fig. 3(d), while the AWGN case suggests an SNR requirement of 13.85 dB, the nominal, nonlinear, and jitter-dominated scenarios require stricter SNR thresholds of 13.90 dB, 13.95 dB, and 14.25 dB, respectively. Consequently, static threshold-based SNR budget predictions incur errors ranging from 0.05 dB to 0.4 dB. Such discrepancies could lead to unforeseen link failures in high-capacity networks; however, our proposed distribution-aware framework successfully captures these non-stationary effects, ensuring accurate reliability monitoring.

Conclusion

Static pre-FEC thresholds fail to ensure link reliability under non-stationary conditions. In this work, we proposed a DTF framework that accurately predicts post-FEC performance by mapping pre-FEC BER distributions. Experimental validation reveals that traditional mean-BER metrics can incur significant link margin errors due to heavy-tailed distributions. By monitoring the γ -th moment, our low-complexity method enables robust and proactive reliability monitoring in autonomous optical networks.

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